

Finite mixture models

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Categorical distribution

Let $Z \sim \text{Cat}(H, p)$ represent a categorical distribution with

- $P(Z = h) = p_h$ for $h = 1, \dots, H$ and
- $\sum_{h=1}^H p_h = 1$.

Example: discrete choice model

Suppose we have a set of H categories and we label these $1, \dots, H$. Independently, consumers choose one of the H categories with the same probability. Then a reasonable model is $Z_i \stackrel{\text{ind}}{\sim} \text{Cat}(H, p)$.

Multinomial distribution

If we count the number of times consumers choose each category, i.e.

$$Y_h = \sum_{i=1}^n \mathbf{I}(Z_i = h),$$

then the result is the multinomial distribution, i.e. $Y \sim \text{Mult}(n, p)$. The multinomial distribution has probability mass function

$$p(y; n, p) = \frac{n!}{y_1! \cdots y_H!} \prod_{h=1}^H p_h^{y_h}$$

which has

- $E[Y_i] = np_i$,
- $V[Y_i] = np_i(1 - p_i)$, and
- $\text{Cov}[Y_i, Y_j] = -np_i p_j$ for $(i \neq j)$.

A special case is $H = 2$ which is the binomial distribution.

Dirichlet distribution

The Dirichlet distribution (named after Peter Gustav Lejeune Dirichlet), i.e. $P \sim \text{Dir}(a)$, is a probability distribution for a probability vector of length H . The probability density function for the Dirichlet distribution is

$$p(P; a) = \frac{\Gamma(a_1 + \cdots + a_H)}{\Gamma(a_1) \cdots \Gamma(a_H)} \prod_{h=1}^H p_h^{a_h-1}$$

where $p_h \geq 0$, $\sum_{h=1}^H p_h = 1$, and concentration parameters $a_h > 0$.

Letting $a_0 = \sum_{h=1}^H a_h$, then some moments are

- $E[p_h] = \frac{a_h}{a_0}$,
- $V[p_h] = \frac{a_h(a_0 - a_h)}{a_0^2(a_0 + 1)}$,
- $\text{Cov}(p_h, p_k) = -\frac{a_h a_k}{a_0^2(a_0 + 1)}$, and
- $\text{mode}(p_h) = \frac{a_h - 1}{a_0 - H}$ for $a_h > 1$.

A special case is $H = 2$ which is the beta distribution.

Conjugate prior for multinomial distribution

The Dirichlet distribution is the natural conjugate prior for the multinomial distribution. If

$$Y \sim Mult(n, \pi) \quad \text{and} \quad \pi \sim Dir(a)$$

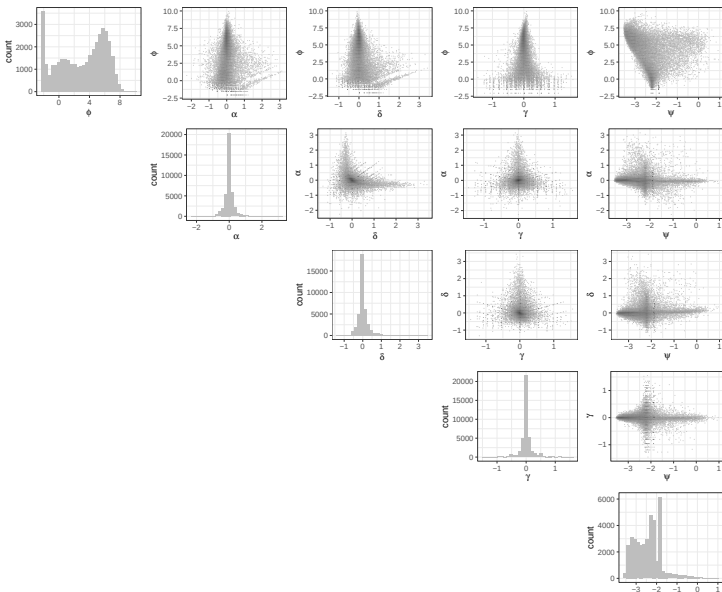
then

$$\pi|y \sim Dir(a + y).$$

Some possible default priors are

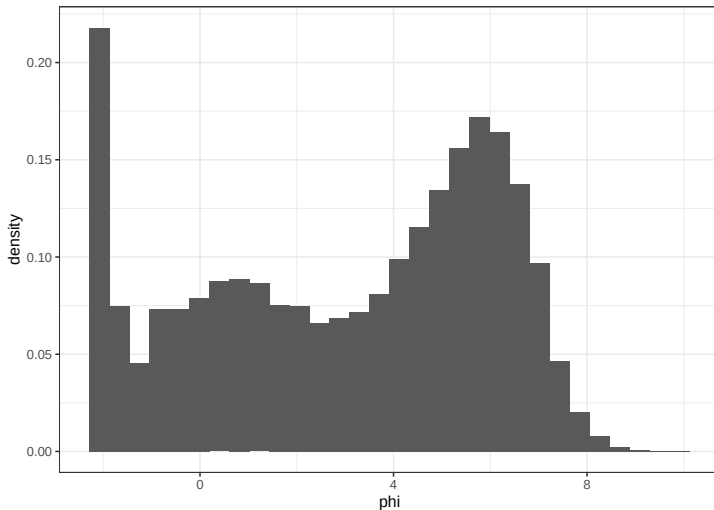
- $a = 1$ which is the uniform density over π ,
- $a = 1/2$ is Jeffreys prior for the multinomial,
- $a = 1/H$, and
- $a = 0$, an improper prior that is uniform on $\log(\pi_h)$. The resulting posterior is proper if $y_h > 0$ for all h .

Complicated distributions



Finite mixtures

Let's focus on modeling the univariate distribution for $\hat{\phi}$



Finite mixture

A finite mixture model for the marginal distribution for $Y_i = \hat{\phi}_i$ is

$$Y_i \stackrel{\text{ind}}{\sim} \sum_{h=1}^H \pi_h N(\mu_h, \sigma_h^2)$$

where $\sum_{h=1}^H \pi_h = 1$.

Alternatively, we can introduce a latent variable $\zeta_i = h$ if observation i came from group h . Then

$$\begin{aligned} Y_i | \zeta_i = z &\stackrel{\text{ind}}{\sim} N(\mu_z, \sigma_z^2) \\ \zeta_i &\stackrel{\text{ind}}{\sim} \text{Cat}(H, \pi) \end{aligned}$$

where $\zeta \sim \text{Cat}(H, \pi)$ is a categorical random variable with $P(\zeta = h) = \pi_h$ for $h = 1, \dots, H$ and $\pi = (\pi_1, \dots, \pi_H)$.

A possible prior

Let's assume

$$\begin{aligned}\pi &\sim \text{Dir}(a) \\ \mu_h | \sigma_h^2 &\stackrel{\text{ind}}{\sim} N(m_h, v_h^2 \sigma_h^2) \\ \sigma_h^2 &\stackrel{\text{ind}}{\sim} IG(c_h, d_h)\end{aligned}$$

and π is independent of $\mu = (\mu_1, \dots, \mu_H)$ and $\sigma^2 = (\sigma_1^2, \dots, \sigma_H^2)$.

Commonly, we have $m_h = m$, $v_h = v$, $c_h = c$, and $d_h = d$. If the data have been standardized (scaled and centered), a reasonable default prior is

- $m = 0$,
- $v = 1$,
- $c = 2$,
- $d = 4$, (BDA3 pg 535) and
- a is $1/H$ (BDA3 pg 536).

MCMC

The steps of a Gibbs sampler with stationary distribution

$$p(\pi, \mu, \sigma^2, \zeta | y) \propto p(y | \zeta, \mu, \sigma^2) p(\zeta | \pi) p(\mu | \sigma^2) p(\sigma^2) p(\pi)$$

has steps

1. For $i = 1, \dots, n$, sample ζ_i from its full conditional (as the ζ are conditionally independent across i):

$$P(\zeta_i = h | \dots) \propto \pi_h N(y_i; \mu_h, \sigma_h^2)$$

2. Jointly sample π and μ, σ^2 as they are conditionally independent.

- a. Sample $\pi \sim \text{Dir}(a + Z)$ where $Z = (Z_1, \dots, Z_H)$ and $Z_h = \sum_{i=1}^n \mathbb{I}(\zeta_i = h)$.
- b. For $h = 1, \dots, H$, sample μ_h, σ_h^2 from their full conditional (as these are conditionally independent across h):

$$\sigma_h^2 \stackrel{\text{ind}}{\sim} IG(c'_h, d'_h) \quad \mu_h | \sigma_h^2 \stackrel{\text{ind}}{\sim} N(m'_h, v_h'^2 \sigma_h^2)$$

where

$$\begin{aligned} v_h'^2 &= (1/v_h^2 + Z_h)^{-1} \\ m'_h &= v_h'^2 (m_h/v_h^2 + Z_h \bar{y}_h) \\ c_h &= c_d + Z_h/2 \\ d'_h &= d_h + \frac{1}{2} \left(\sum_{i:\zeta_i=h} (y_i - \bar{y}_h)^2 + \left(\frac{Z_h}{1 + Z_h/v_h^2} \right) (\bar{y}_h - m_h)^2 \right) \\ \bar{y}_h &= \frac{1}{Z_h} \sum_{i:\zeta_i=h} y_i \end{aligned}$$

```
library("rjags")
jags_model = "
model {
  for (i in 1:n) {
    y[i] ~ dnorm(mu[zeta[i]], tau[zeta[i]])
    zeta[i] ~ dcat(pi[])
  }

  for (i in 1:H) {
    mu[i] ~ dnorm(0,1e-5)
    tau[i] ~ dgamma(1,1)
    sigma[i] <- 1/sqrt(tau[i])
  }

  pi ~ ddirich(a)
}"
```

```
tmp = hat[sample(nrow(hat), 1000),]
dat = list(n = nrow(tmp), H = 3, y = tmp$phi, a = rep(1,3))
```

```
jm = jags.model(textConnection(jags_model), data = dat, n.chains = 3)
r = coda.samples(jm,
  variable.names = c('mu','sigma','pi'),
  n.iter = 1e4)
```

Convergence diagnostics - Potential Scale Reduction Factor

```
gelman.diag(r)
```

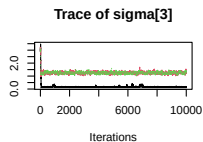
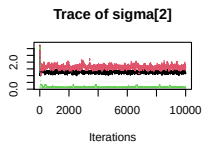
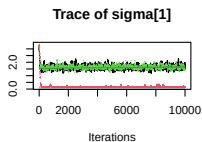
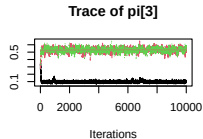
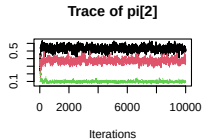
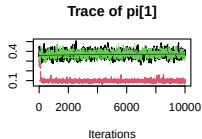
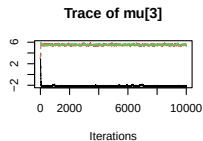
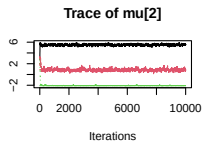
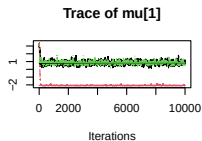
```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## mu[1]      14.1      33.6
## mu[2]      40.9     122.4
## mu[3]      69.8     158.5
## pi[1]       9.0     20.0
## pi[2]      12.8     28.9
## pi[3]      14.5     32.4
## sigma[1]   10.8     25.1
## sigma[2]   12.4     33.7
## sigma[3]   14.0     29.7
##
## Multivariate psrf
##
## 95.2
```

```
gelman.diag(r, multivariate=FALSE)
```

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## mu[1]      14.1      33.6
## mu[2]      40.9     122.4
## mu[3]      69.8     158.5
## pi[1]       9.0     20.0
## pi[2]      12.8     28.9
## pi[3]      14.5     32.4
```

Convergence diagnostics - Traceplots

```
plot(r, density=FALSE)
```



Prior distributions

The parameters of the model are unidentified due to **label-switching**, i.e.

$$Y_i \stackrel{ind}{\sim} \sum_{h=1}^H \pi_h N(\mu_h, \sigma_h^2) \stackrel{d}{=} \sum_{h'=1}^H \pi_{h'} N(\mu_{h'}, \sigma_{h'}^2)$$

for some permutation h' .

One way to resolve this issue is to enforce identifiability in the prior. For example, in one-dimension, we can order the component means:

$$\mu_1 < \mu_2 < \cdots < \mu_H.$$

To ensure the posterior is proper

- Maintain proper prior for π
- Ensure proper prior for ratios of variances
(perhaps by ensuring prior is proper for variances themselves)

Two conditionally conjugate prior options

Option 1:

$$Dir(\pi; a)I(\mu_1 < \cdots < \mu_H) \prod_{h=1}^H N(\mu_h; m_h, v_h^2) IG(\sigma_h^2; c_h, d_h)$$

Option 2:

$$Dir(\pi; a)I(\mu_1 < \cdots < \mu_H) \prod_{h=1}^H N(\mu_h; m_h, v_h^2 \sigma_h^2) IG(\sigma_h^2; c_h, d_h)$$

```
library("rjags")
jags_model = "
model {
  for (i in 1:n) {
    y[i] ~ dnorm(mu[zeta[i]], tau[zeta[i]])
    zeta[i] ~ dcat(pi[])
  }

  for (i in 1:H) {
    mu0[i] ~ dnorm(0,1e-5)
    tau[i] ~ dgamma(1,1)
    sigma[i] <- 1/sqrt(tau[i])
  }

  mu[1:H] <- sort(mu0)      # sort the means
  pi ~ ddirich(a)
}"
```

```
jm = jags.model(textConnection(jags_model), data = dat, n.chains = 3)
r = coda.samples(jm,
  variable.names = c('mu', 'sigma', 'pi', 'zeta'),
  n.iter = 1e4,
  thin = 10)
```


Convergence diagnostics

```

gelman.diag(r)

## Error in chol.default(W): the leading minor of order 39 is not positive

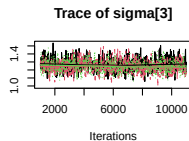
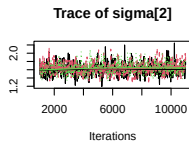
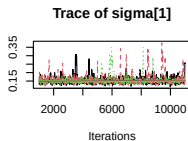
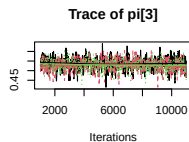
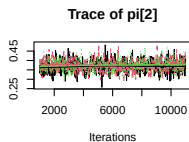
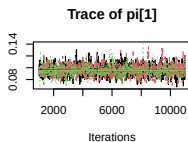
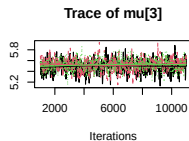
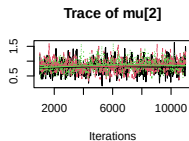
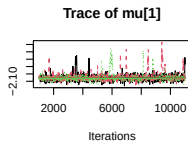
gelman.diag(r, multivariate=FALSE)

## Potential scale reduction factors:
##
##           Point est. Upper C.I.
## mu[1]          1.053      1.079
## mu[2]          1.015      1.054
## mu[3]          1.014      1.048
## pi[1]          1.007      1.010
## pi[2]          1.009      1.035
## pi[3]          1.012      1.042
## sigma[1]       1.064      1.108
## sigma[2]       1.008      1.029
## sigma[3]       1.022      1.078
## zeta[1]        1.039      1.041
## zeta[2]        1.018      1.026
## zeta[3]        1.002      1.002
## zeta[4]        1.019      1.034
## zeta[5]        1.041      1.053
## zeta[6]        1.005      1.009
## zeta[7]        0.999      0.999
## zeta[8]        1.020      1.022
## zeta[9]        1.121      1.149
## zeta[10]       1.105      1.108
## zeta[11]       1.039      1.041
## zeta[12]       1.002      1.008

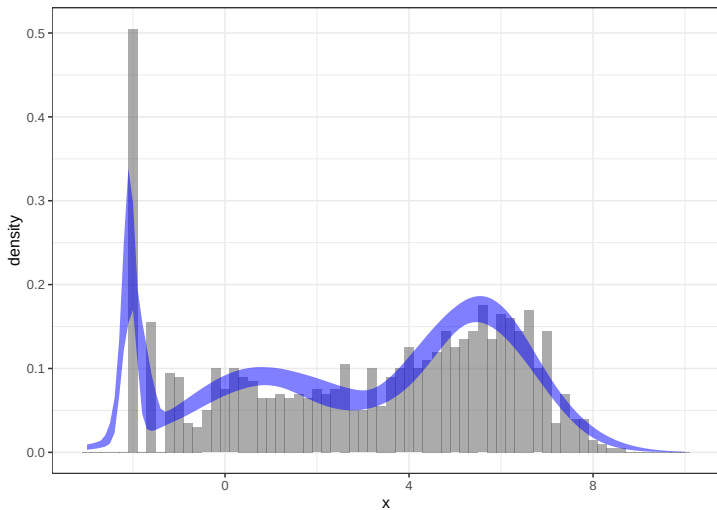
```

Convergence diagnostics (2)

```
plot(r, density=FALSE)
```



Posterior on data density



Group membership

Group membership can be obtained using the ζ_i , e.g.

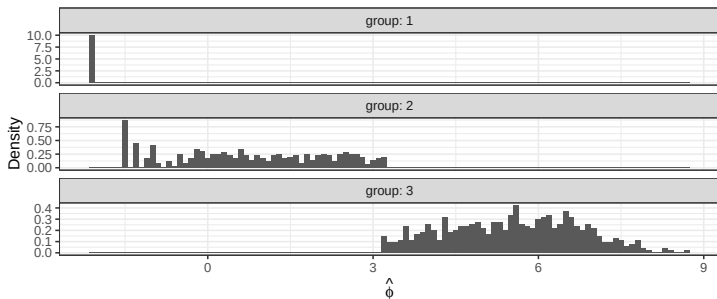
$$P(\text{gene } i \text{ in cluster } h) = P(\zeta_i = h|y) \approx \sum_{m=1}^M \mathbb{I}(\zeta_i^{(m)} = h).$$

```
## # A tibble: 6 x 4
##   parameter    p1      p2      p3
##   <int> <dbl> <dbl> <dbl>
## 1         1  0     0.004  0.996
## 2         2 0.033  0.967    0
## 3         3  0     0.01   0.99
## 4         4 0.934  0.0657  0
## 5         5  0     0.981  0.0193
## 6         6  0     0.0837  0.916
```

Clustering

Genes can then be clustered by assigning them to a group based on their posterior probabilities of group membership, i.e. for gene i , we assign the group according to

$$\operatorname{argmax}_h P(\zeta_i = h | y).$$



Clustering is extremely sensitive to the parametric model chosen, e.g. normal in this example, and the cluster could change dramatically with a different choice, e.g. t .

Choosing H

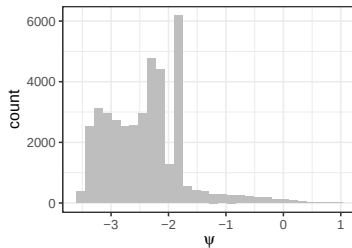
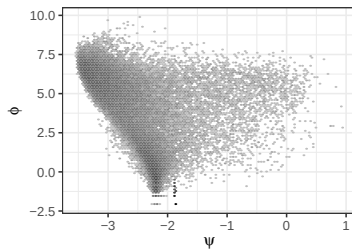
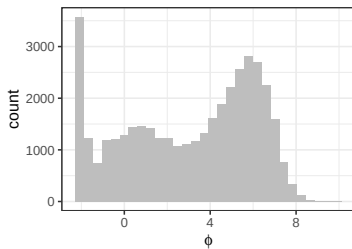
When using finite mixture models one of the key choices is to choose H , the number of clusters.

- A Bayesian approach would place a prior on H , e.g. a Poisson or truncated Poisson, and then use reversible jump MCMC to estimate it.
- A more pragmatic approach is to start with a small H and then determine whether there is some feature of the data that is not being adequately addressed, e.g. via posterior predictive pvalues.
- An empirical Bayes finds an MLE (or MAP) via

$$\hat{H} = \operatorname{argmax}_H p(y|H) = \int p(y|\pi, \mu, \sigma^2, H) p(\pi, \mu, \sigma^2|H) d\pi d\mu d\sigma^2$$

and then condition on $\hat{H}$ in the analysis. Typically this MLE (or MAP) is found via the EM algorithm.

Multivariate density estimation



Finite mixture

A model for the joint distribution for $Y_i = (\phi_i, \psi_i)^\top$ is

$$Y_i \stackrel{\text{ind}}{\sim} \sum_{h=1}^H \pi_h N(\mu_h, \Sigma_h)$$

where $\sum_{h=1}^H \pi_h = 1$.

Alternatively, we can introduce a latent variable $\zeta_i = h$ if observation i came from group h . Then

$$\begin{aligned} Y_i | \zeta_i = z &\stackrel{\text{ind}}{\sim} N(\mu_z, \Sigma_z) \\ \zeta_i &\stackrel{\text{ind}}{\sim} \text{Cat}(H, \pi) \end{aligned}$$

where $\zeta \sim \text{Cat}(H, \pi)$ is a categorical random variable with $P(\zeta = h) = \pi_h$ for $h = 1, \dots, H$ and $\pi = (\pi_1, \dots, \pi_H)$.

A possible prior

Let's assume

$$\begin{aligned}\pi &\sim \text{Dir}(a) \\ \mu_h | \Sigma_h &\overset{\text{ind}}{\sim} N_p(\mu_h, v_h^2 \Sigma_h) \\ \Sigma_h &\overset{\text{ind}}{\sim} IW(D_h, c_h)\end{aligned}$$

where $c_h > p - 1$ is the degrees of freedom and D is the scale matrix. The mean of this distribution is $D_h / (c_h - p - 1)$ for $c_h > p + 1$.

MCMC

The steps of a Gibbs sampler with stationary distribution

$$p(\pi, \mu, \Sigma, \zeta|y) \propto p(y|\zeta, \mu, \Sigma)p(\zeta|\pi)p(\mu|\Sigma)p(\Sigma)p(\pi)$$

has steps

1. For $i = 1, \dots, n$, sample ζ_i from its full conditional

$$P(\zeta_i = h | \dots) \propto \pi_h N(y_i; \mu_h, \Sigma_h)$$

2. Jointly sample π and μ, σ^2 as they are conditionally independent.

- a. Sample $\pi \sim Dir(a + Z)$ where $Z = (Z_1, \dots, Z_H)$ and $Z_h = \sum_{i=1}^n I(\zeta_i = h)$.
- b. For $h = 1, \dots, H$, sample μ_h, Σ_h from their full conditional

$$\Sigma_h \stackrel{ind}{\sim} IW(D'_h, c'_h) \quad \mu_h | \Sigma_h \stackrel{ind}{\sim} N(m'_h, v_h'^2 \Sigma_h)$$

where

$$\begin{aligned} v_h'^2 &= (1/v_h^2 + Z_h)^{-1} \\ m'_h &= v_h'^2 (m_h/v_h^2 + Z_h \bar{y}_h) \\ c_h &= c_d + Z_h \\ D'_h &= D_h + \sum_{i:\zeta_i=h} (y_i - \bar{y}_h)(y_i - \bar{y}_h)^\top + \left(\frac{Z_h}{1 + Z_h/v_h^2} \right) (\bar{y}_h - \mu_h)(\bar{y}_h - \mu_h)^\top \\ \bar{y}_h &= \frac{1}{Z_h} \sum_{i:\zeta_i=h} y_i \end{aligned}$$

```
library("rjags")
joint_mixture_model = "
model {
  for (i in 1:n) {
    y[i,1:p] ~ dnmnorm(mu[,zeta[i]], Tau[,zeta[i]])
    zeta[i] ~ dcat(pi[])
  }

  for (h in 1:H) {
    mu[1:p,h] ~ dnmnorm(mu0,Tau[,h])
    Tau[1:p,1:p,h] ~ dwish(D[,c],c)
    Sigma[1:p,1:p,h] <- inverse(Tau[,h])
  }

  pi ~ ddirich(a)
}"
```

```
tmp = hat[sample(nrow(hat), 1000),]
dat = list(n=nrow(tmp), y = tmp[,c('phi','psi')], p=2, H=10)
dat$a = rep(1/dat$H, dat$H)
dat$D = diag(1, dat$p)
dat$c = dat$p+1
dat$mu0 = c(3,0)
```

```
jm = jags.model(textConnection(joint_mixture_model),
  data = dat,
  n.chains = 3)
r = coda.samples(jm, variable.names = c('pi','mu','Sigma'), n.iter = 1e4)
```